

ROUND: keep the promise specific

A small check can earn a useful amount of trust

01 / THE SAS EXAMPLE

```
data test;
  input value;
  rounded = round(value);
  datalines;
2.3
2.7
-1.4
-1.6
3.0
;
run;

proc print data=test noobs;
run;
```

REPORTED TEST SETUP / 21 SEP 2026
Claude Sonnet 4.5 / AWS Bedrock GovCloud
Claude Code 2.1.278 / effortLevel=high / prompt v0.1.0

ATTACH SOURCE OR PASTE IT AFTER THE PROMPT. START FRESH EACH TIME.

WHAT YOU WILL PRACTICE

Match a simple conversion to an explicit scope and a compact result check.

Round five positive and negative numbers to the nearest integer. None is a halfway value. This exercise establishes agreement only for these supplied values.

EXPECTED RESULT

Row	value	rounded
1	2.3	2
2	2.7	3
3	-1.4	-1
4	-1.6	-2
5	3.0	3

Five non-halfway inputs only; numeric 2 equals 2.0.

REPORTED NUMERIC MATCHES / FIVE INPUTS
Prompt A 3/3 | Prompt B 3/3

Reference values checked in Python; SAS execution pending.

Prompt A: leave choices open

High degrees of freedom / Notice the decisions the model makes for you.

COPY THIS PROMPT + THE SAS EXAMPLE

Convert this SAS code to Python.

Show code in this response. Do not create or run files, or read existing Python files.

REPORTED TRIAL RESULTS / 21 SEP 2026

Matching numbers, conflicting explanations

Run	Method	Values	Explanation
A1	.round()	Yes	Wrong SAS rounding claim
A2	.round()	Yes	Correct + alternative recipe
A3	.round()	Yes	Correct + scope note

A1 wrongly said SAS and pandas both use ties-to-even.

Correct output and correct explanation are separate checks.

All A runs matched [2, 3, -1, -2, 3].

SOURCES: results-20260921.csv; trial-log-20260921.md; analysis-20260921.md

WHY CHOICES REMAIN OPEN

- 1

A1's explanation was wrong

At exact halfway values, SAS ROUND rounds away from zero; pandas rounds to even. These inputs do not test that difference.
- 2

A2 and A3 recognized the difference

A2 offered a floor/ceil alternative. A3 correctly explained that this fixture has no halfway values.
- 3

Keep the conclusion within scope

Five matching outputs do not establish equivalence for all SAS ROUND inputs or other rounding units.

ANTI-PATTERN: ACCEPTING OUTPUT AS PROOF

Review the explanation as well as the values. A correct result on this fixture cannot validate an untested general claim.

Expand the checks when the required scope expands.

Prompt B: make the choices explicit

Low degrees of freedom / State the behavior, then ask for a check.

COPY THIS PROMPT + THE SAME SAS EXAMPLE

```
Convert this SAS code to Python using pandas.
- Build a DataFrame with value in the original row order:
  [2.3, 2.7, -1.4, -1.6, 3.0].
- Create rounded using the value Series .round(0) method.
- Print value and rounded. Numeric 2 and 2.0 are equivalent here.
- Include an assertion that rounded values are [2, 3, -1, -2, 3].
- Briefly explain what the check covers: these five inputs,
  nearest-integer rounding, including the negative inputs.
- State that halfway values and other rounding units have not
  been tested. Keep those cases outside this exercise.
```

Show code in this response. Do not create or run files, or read existing Python files.

WHY THESE DETAILS HELP

- 1 Choose one route**
Name pandas and .round(0). All three B runs report the requested method.
- 2 Define numeric agreement**
Check the list [2, 3, -1, -2, 3] in order. Numeric 2 and 2.0 count as equal here.
- 3 State the limits of the check**
B1-B3 reportedly included the requested scope note: halfway inputs and other rounding units were not tested.

REPORTED RESULTS / THREE B RUNS

Run	Numeric match	Scope note
B1	Yes	Included
B2	Yes	Included
B3	Yes	Included

CHECK IT YOURSELF
Assert all five rounded values in order.
Keep inputs unchanged; record the scope of the pass.

One behavior, three implementations

The same five non-halfway values round to the same integers. These translations calculate a result column while retaining the input values.

SAS / SOURCE BEHAVIOR

```
%macro transform(in=, out=);
data &out;
  set &in;
  rounded = round(value);
run;
%mend;
```

ROUND uses unit 1 by default. The macro reads a SAS dataset and writes a new one. SAS harness is supplied but unexecuted.

PYTHON / PANDAS

```
def translate(df):
    out = df.copy(deep=True)
    out["rounded"] = out["value"].round(0)
    return out
```

pandas Series.round(0) computes nearest integers on a copied DataFrame. Numeric displays such as 2 and 2.0 are equivalent here.

R / BASE

```
translate <- function(df) {
  out <- df
  out$rounded <- round(out$value, 0)
  out
}
```

Base R round(value, 0) returns numeric results. Its halfway behavior must not be generalized to SAS ROUND.

Keep the scope explicit

The original five values and the added integer-boundary probes exclude halfway cases. Other units and SAS floating-point adjustments are outside scope.

Output and explanation differ

Matching results on these rows do not prove all ROUND implementations equivalent. pandas and R use ties-to-even; SAS documents a different exact-tie rule.

SAS ROUND

pandas Series.round

R round

Three checks that earn the pass

Use the same inputs and expected behavior in each language. Check the contract as well as the values.

T1 / Original fixture: known answers

INPUT / CONDITION

Original five values: 2.3, 2.7, -1.4, -1.6, 3.0.

EXPECTED

rounded = [2, 3, -1, -2, 3].

WHAT THIS CATCHES

Wrong direction, truncation, missed negative values or hard-coded mismatched outputs.

T2 / Added probes around integers

INPUT / CONDITION

Added validation: -2.01, -1.99, -0.01, 0.01, 1.99, 2.01. No halfway values.

EXPECTED

rounded = [-2, -2, 0, 0, 2, 2].

WHAT THIS CATCHES

Floor, ceiling or truncation substituted for nearest-integer rounding.

T3 / Preserve inputs, types and order

INPUT / CONDITION

All CSV rows in reverse order, with custom row identifiers.

EXPECTED

Fresh result; source values unchanged; original order retained; rounded numeric; reordered expected values match.

WHAT THIS CATCHES

Input mutation, sorting, result text instead of numbers, or position-dependent/hard-coded answers.

REFERENCE CODE + TESTS: `module-05-round/appendix/`

Run: `python tools/run_appendix_tests.py --modules 05`

New reference code, separate from historical trials. SAS checks await execution; scope is stated in the companion README.